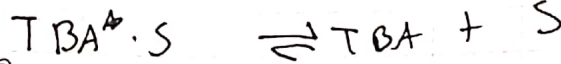
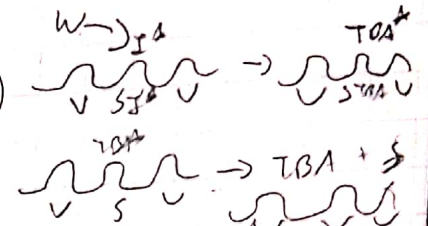
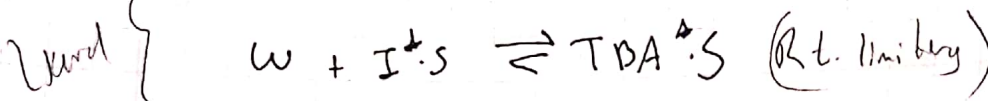
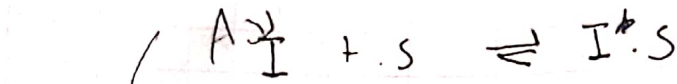
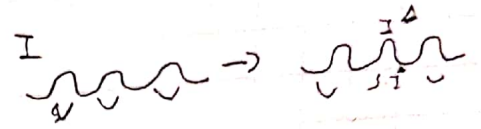


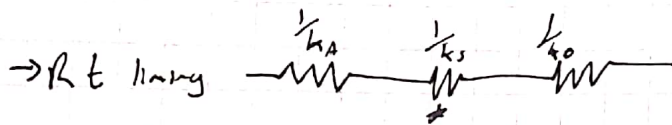
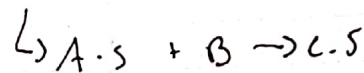
Exam 2

David Alvarado

1) Given rxn mechanism



Derive the rate law assuming the Eley-Rideal kinetics is limiting



$$\Rightarrow -r_A = r_{A0} = r_S = r_D$$

$$k_S \ll k_A, k_D$$

$$\Rightarrow \frac{1}{k_A}, \frac{1}{k_D} \ll \frac{1}{k_S}$$

Rev

$$\Rightarrow \left(\frac{r_{A0}}{k_A} \right) = \frac{r_S}{k_S} = \left(\frac{r_D}{k_D} \right)$$

$$r_S = k_S \left[C_W C_{I \cdot S} - \frac{C_{TBA \cdot S}}{k_S} \right]$$

$$\textcircled{1} \frac{r_{A0}}{k_A} \approx 0$$

$$\frac{1}{k_A} [k_{A0} (C_I C_W - \frac{C_{I \cdot S}}{k_A})] = 0$$

$$C_I C_W = \frac{C_{I \cdot S}}{k_A}$$

$$C_{I \cdot S} = k_A C_I C_W$$

$$\textcircled{2} \frac{r_D}{k_D} \approx 0$$

$$\frac{1}{k_D} [k_D (C_{TBA \cdot S} - k_D C_{TBA} C_W)] = 0$$

$$C_{TBA \cdot S} = k_D C_{TBA} C_W$$

$$C_{TBA \cdot S} = k_D C_{TBA} C_W$$

Site Balance

$$C_T = C_W + C_{I \cdot S} + C_{TBA}$$

$$C_T = C_W + k_A C_I C_W + k_D C_{TBA} C_W$$

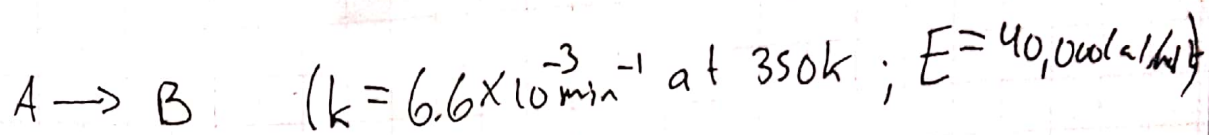
$$C_W = \frac{C_T}{1 + k_A C_I + k_D C_{TBA}}$$

$$-r_A = k_s \left[C_w \underbrace{C_I}_{I_{ss}} - \frac{C_{TBA,s}}{K_s} \right]$$

$$= k_s \left[C_w \left(k_A C_I \frac{C_T}{1 + k_A C_I + k_D C_{TBA}} \right) - \frac{1}{K_s} \left(k_D C_{TBA} \frac{C_T}{1 + k_A C_I + k_D C_{TBA}} \right) \right]$$

$$= \frac{C_T \cdot k_s \left[C_w k_A C_I - \frac{k_D}{K_s} C_{TBA} \right]}{1 + k_A C_I + k_D C_{TBA}}$$

2) First-order irrev. exothermic liq phase rxn

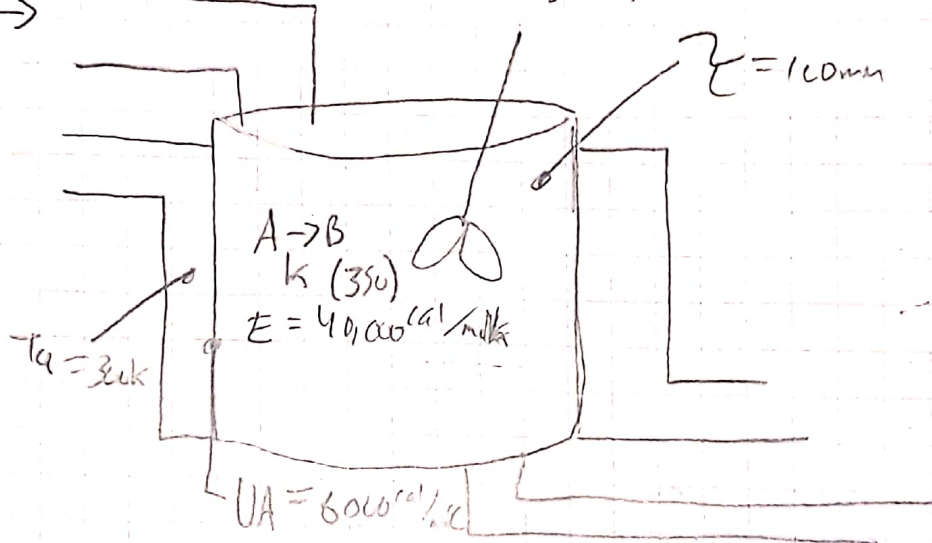


is to be carried out in a jacketed CSTR ($UA = 8000 \frac{\text{cal}}{\text{min}^\circ\text{C}}; T_a = 300\text{K}$)
Species A & an inert feed to rxn in equimolar amts.

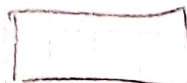
The molar flow rate of A is $80 \frac{\text{mol}}{\text{min}}$ & $\tau = 100 \text{ min}$

Heat capacity; $C_{p,A} = 20 \frac{\text{cal}}{\text{mol}^\circ\text{C}}$
 $C_{p,B} = 20 \frac{\text{cal}}{\text{mol}^\circ\text{C}}$
 $C_{p,I} = 30 \frac{\text{cal}}{\text{mol}^\circ\text{C}}$ (constant) $\Delta H_{rxn} = -7500 \frac{\text{cal}}{\text{mol}}$

$$F_A = F_I = 80 \frac{\text{mol}}{\text{min}}$$



a) Find reactor temp if $T_{feed} = 450\text{K}$



$$\Delta C_p = C_{p,A} - C_{p,B} = 0$$

$$\underbrace{\dot{Q} - F_{A0} \sum_{i=1}^N \theta_i C_{p,i} dT}_{(1)} - \underbrace{F_{A0} \times [\Delta H_{rxn}(T_R)]}_{(2)} + \underbrace{\int_{T_R=0}^T \Delta C_p dT}_{(3)} = 0$$

$$(1) \quad \dot{Q} = UA(T_a - T)$$

$$(2) \quad -F_{A0} \left(\theta_A C_{p,A} \left(\int_{T_0}^T dT \right) + \theta_I C_{p,I} \left(\int_{T_0}^T dT \right) \right) - F_{A0} ((C_{p,A} + C_{p,I})(T - T_0))$$

$$(3) \quad -F_{A0} \tau \Delta H_{rxn}$$

$$\theta_i = \frac{F_{i0}}{F_{A0}}$$

$$\theta_A = \frac{F_{A0}}{F_{A0}} = 1$$

$$\theta_I = \frac{F_{I0}}{F_{A0}} = 1$$

③ $\frac{mol}{min} - F_{A0} X \Delta H_{rxn}$

Break $\Rightarrow X = \frac{h \tau}{1 + h \tau}$

$\rightarrow k = f(T)$

Note $\left(\frac{k}{k_0} = \frac{\exp(-E/RT)}{\exp(-E/RT_0)} \right)$

$k(T) = k(T_0) \exp\left(\frac{E}{R} \left(\frac{1}{T_0} - \frac{1}{T}\right)\right)$

$\Rightarrow X = \frac{k(T_0) \exp\left(\frac{E}{R} \left(\frac{1}{T_0} - \frac{1}{T}\right)\right) \tau}{1 + k(T_0) \exp\left(\frac{E}{R} \left(\frac{1}{T_0} - \frac{1}{T}\right)\right) \tau}$ AAA

All together

$\Rightarrow UA(T_c - T) - F_{A0} (C_{PA} + C_{PX})(T - T_0) - F_{A0} X \Delta H_{rxn} = 0$

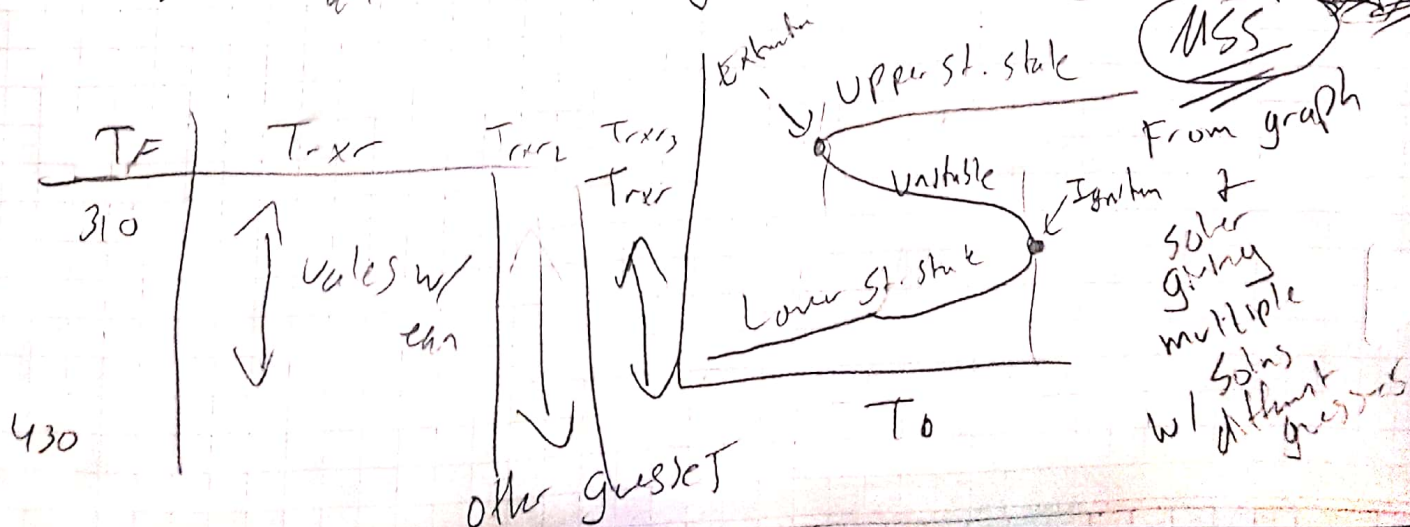
$800 \frac{cal}{min} (300K - T) - 80 \frac{mol}{min} (50 \frac{cal}{mol} (T - 450K)) - 80 \frac{mol}{min} (-7500 \frac{cal}{mol}) X = 0$

Soe for T very excell goal seek

$T = 399.9425K$

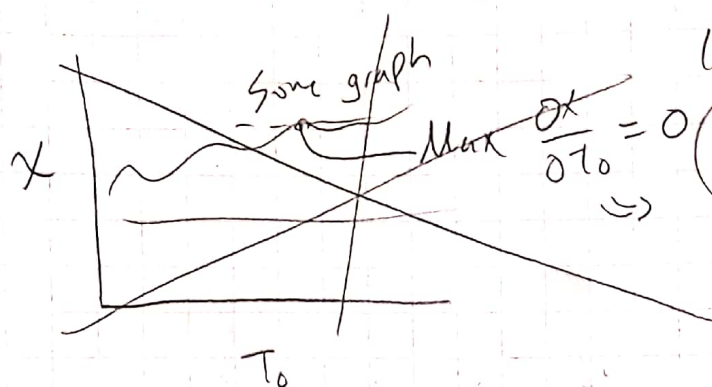
b) Plot & Analyze reactor temp w/ (310K to 430K)

Same eqn as (a) just switch $T_0 = T_{feed}$ AAA



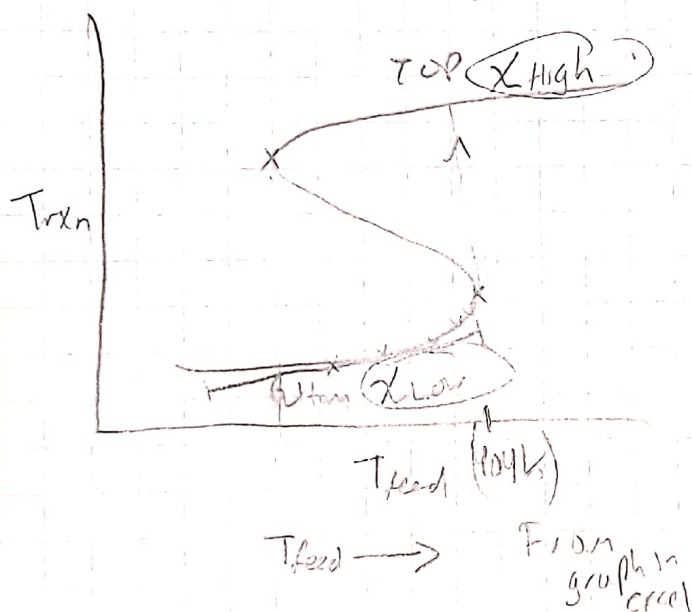
c) What inlet temp. must the fluid be preheated for reactor at a high conversion?

What are the corresponding temp & concn of fluid in CSTR on inlet temp?



Consider

MSS multiple
conversions



T preheated?

Temp	X_{low}	X_{top}
↑	↑	↑
↓	↓	↓
	0.000944	0.99
	Low	

Jump in conversion at Ignition temp. $T = 404K$

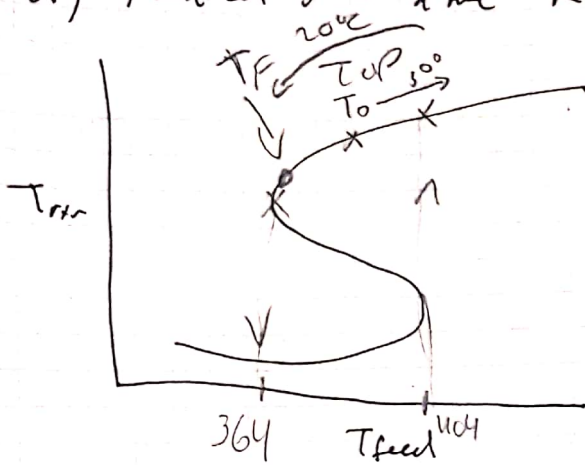
$$\Rightarrow [T_0 = 404K]$$

From $X = \frac{k_c \tau}{1 + k_c \tau}$ Excel

TOP

$$\left[\begin{array}{l} X = 0.990595 \\ T_{rxn} = 383.855 \text{ (2 solns)} \end{array} \right], 394.74$$

d) Heated sul above Reator temp + cooled 20°C



$$T_{rxr}(C) = 383.855$$

→ High X

⇒ T_{OP}

$$383.855 - 20$$

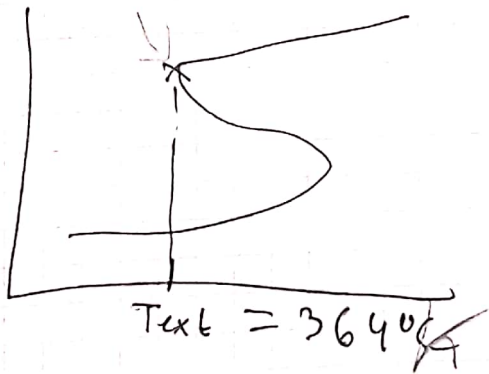
$$= 363.855$$

$$T > T_{exchanger}$$

Excel

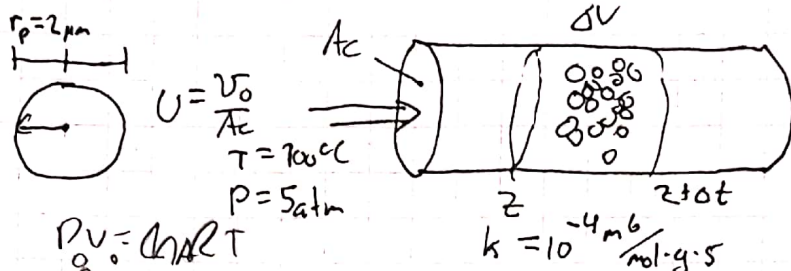
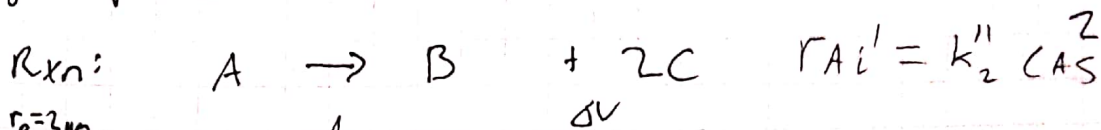
$$At \quad T = 363.855 \Rightarrow \quad \boxed{X = 0.9274}$$

e) Inlet exchanger temp



$$\Rightarrow T_{ex} = 364 K$$

3) Second Order Rxn is taking place in a micro PBR w/ micro-sized catalytic particles:



$$PV = nRT$$

$$\frac{P_0}{nT_0} = C_{A0}$$

$$C_{A0} = \frac{5 \text{ atm}}{8.2 \times 10^{-5} \frac{\text{m}^3 \text{ mol}^{-1} \text{ K}^{-1}}{\text{K}} \cdot 373.15 \text{ K}}$$

$$D = 2.66 \times 10^{-9} \text{ m}^2/\text{s} \quad (C_{A0} = 163.407 \text{ mol/m}^3) \quad X = 0.99$$

Porosity = 0.4 = ϕ

Cat. density = $100 \text{ g/m}^3 = \rho_c$

Internal SA = $0.05 \text{ m}^2/\text{g}$

PBR

$$\frac{dF_A}{dw} = r_{A1}' a_c L_{p,c} \quad F_A = F_{A0}(1-X)$$

$$\frac{dF_A}{dw} = \int_0^L (r_{A1}') \quad F_A = C_A \cdot v$$

$$\textcircled{1} \frac{dX}{dw} \rightarrow \frac{dX}{dz}$$

$$W = V_c \rho_c$$

$$W = [(1-\epsilon) A_c z] \rho_c$$

$$dw = (1-\epsilon) A_c \rho_c dz$$

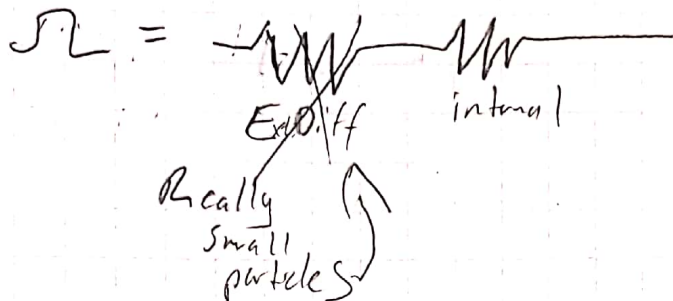
$$\frac{1}{dw} = \frac{1}{dz} \frac{1}{(1-\epsilon) A_c \rho_c}$$

$$\left[\frac{dX}{dw} = \frac{dX}{dz} \cdot \frac{1}{(1-\epsilon) A_c \rho_c} \right]$$

$$\textcircled{2} F_A = F_{A0}(1-X)$$

$$③ \quad r'_{\text{Actual}} = \underbrace{\Omega}_{\text{Break}} \underbrace{(r_A')_i}_{\text{2nd order?}}$$

$$\text{Break } \Omega = ? \quad (\text{2nd order?})$$



$$\begin{aligned}
 r_A' &= \underbrace{k''}_{k'} C_A S^2 \\
 &= \underbrace{k''}_{k'} (C_{A0}(1-x))^2 \\
 &= \underbrace{k''}_{k'} (C_{A0}(1-x))^2
 \end{aligned}$$

$$\Omega = \eta \quad (\text{2nd order})$$

$$\begin{aligned}
 \eta &= \left(\frac{2}{n+1} \right)^{1/2} \frac{3}{\phi_n} \\
 &= \left(\frac{2}{3} \right)^{1/2} \frac{3}{\phi_2}
 \end{aligned}$$

$$n=2$$

$$\phi_2 = R \sqrt{\frac{k_2'' S_{A0} C_{A0}}{D_e}}$$

$$= R \sqrt{\frac{k_1' S_{A0} C_{A0}}{D_e}}$$

$$\phi_2 = 2 \times 10^{-6} \sqrt{\frac{10^{-4} \text{ m}^2/\text{s} \cdot 100 \text{ g/m}^3 \cdot 163.467 \text{ mol/m}^3}{2.66 \times 10^{-4} \text{ m}^2/\text{s}}}$$

$$\boxed{\phi_2 = 0.04957}$$

$$\Rightarrow \eta = \left(\frac{2}{3} \right)^{1/2} \frac{3}{0.04957} = 49.414$$

$$\Rightarrow \frac{dF_A}{dw} = -\Omega(r_A')$$

$$-(F_A) \frac{dx}{dz} \frac{1}{(1-\epsilon)A_c \rho_c} = -\Omega r_A'$$

$$C_{A_0} \frac{dx}{dz} \frac{1}{(1-\epsilon)A_c \rho_c} = -\Omega r_A'$$

$$\frac{C_{A_0} U}{(1-\epsilon) \rho_c} \frac{dx}{dz} = -\Omega r_A'$$

$$\frac{C_{A_0} U}{(1-\epsilon) \rho_c} \frac{dx}{dz} = -\Omega k' [C_{A_0}(1-x)]^2$$

$$\frac{C_{A_0} U}{(1-\epsilon) \rho_c} \frac{dx}{dz} = \Omega k' C_{A_0}^2 (1-x)^2$$

$$\frac{U}{(1-\epsilon) \rho_c} \frac{dx}{dz} = \Omega k' C_{A_0} (1-x)^2$$

$$\int_0^{x_f} \frac{dx}{(1-x)^2} = \int_0^L \frac{\Omega k' C_{A_0} (1-\epsilon) \rho_c dz}{U}$$

$$\left[\frac{1}{1-x} \right]_0^{x_f} = \frac{\Omega k' C_{A_0} (1-\epsilon) \rho_c L}{U}$$

$$\left(\frac{1}{1-x_f} \right) - \frac{1}{1-0} = \frac{\Omega k' C_{A_0} (1-\epsilon) \rho_c L}{U}$$

$$\frac{1}{1-0.99} - 1 = \delta L$$

$$\frac{99}{\delta} = L$$

$$\delta = \frac{(49.414)(10^{-4} \frac{m}{s}) (163.407 \frac{m}{s}) (1-0.4)(100 \frac{g}{m^3})}{(3 \times 10^{-6} \frac{m}{s})}$$

$$(\delta = 16149187 \frac{1}{m})$$

$$\Rightarrow L = \frac{99}{16149187 \frac{1}{m}}$$

$$L = 6.13 \times 10^{-6} m$$

check

$$V_{total}(1-\epsilon) = V_{catalyst}$$

$$V_{total} = \frac{V_{catalyst}}{1-\epsilon}$$

$$V = A_c \cdot L$$

$$\pi r^2$$

V really small?

wrong

Conclusion